

The Practical Zoology for Advanced Level and Intermediate Students is a laboratory manual that covers various zoological experiments. The book presents methods, techniques, and illustrations relevant to zoological experiments. The text first, discusses laboratory regulations, microscopical techniques, and then proceeds to tackling the morphology and anatomy of various animals and vertebrate embryology. The book will be most useful to students for understand the practical procedures and drawing of animals in simple manner.

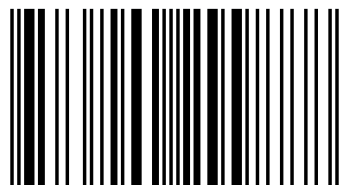


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The practical manual for allied zoology -invertebrates and chordates

Affiliated Periyar University syllabus -2017 onwards



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THE PRACTICAL MANUAL FOR ALLIED ZOOLOGY
INVERTEBRATES AND CHORDATES
(Affiliated Periyar University syllabus -2017 onwards)

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THE PRACTICAL MANNUAL FOR ALLIED ZOOLOGY

GENERAL SAFETY RULES IN THE LABORATORY

1. Determine the location of fire extinguishers, chemical safety showers and eye washers, chemical spill kits and alternative exit routes for lab evacuation.
2. Smoking, eating or drinking in the lab room is totally prohibited.
3. Wear lab aprons when working with chemicals, hot material or preserved specimens.
4. Wear safety goggles when using dangerous chemicals, hot liquids or burners.
5. Any chemicals spilled on the hands or other parts of the skin should be washed off immediately with a plenty of running water.
6. If you have an open skin wound, be sure that it is covered with a water proof bandage.
7. Never work alone in the laboratory.
8. Keep your work area clean and dry.
9. Turn of all electrical equipment, water and gas when it is not in use, especially at the end of the laboratory period.
10. Tie back long hair.
11. Report all chemicals spills or fluids to your instructor immediately for proper clean up.

Special precautions for working with heat or fire:

1. Never leave a lighted Bunsen burner or hot object unattended. When an object is removed from the heat & left to cool, it should be placed where it is shielded from contact.
2. Inflammable liquid bottles should not be left open, not dispensed near a naked flame, hot electric element or electric motor.
3. Use test tube holders to handle hot laboratory equipments.
4. To heating something in a container such as a test tube, always point the open end of the container away from yourself & others.
5. Use only Pyrex glassware's for heating.
6. Allow hot materials to cool before moving them from your lab station.
7. Make sure that Bunsen burner hoses fit tightly.

Special precautions for working with chemicals

1. Never taste or touch substances in the laboratory without specific instructions.
2. Never smell substances in the laboratory without specific instructions.
3. Use materials only from containers that are properly labeled.
4. Wash your hand after working with chemicals.
5. Do not add water to acid. Instead, dilute the acid by adding it to water.
6. Mix heat generating chemicals slowly.

Special precautions for working with electrical equipment

1. Make sure the area under & around the electrical equipment is dry.
2. Never touch electrical equipment with wet hands.
3. Make sure the area surrounding the electrical equipment is free of flammable materials.
4. Turn off all power switches before plugging an appliance into an outlet.

Special Precaution for working with Glassware's and other laboratory equipments

1. Become familiar with the names and appearance of all the laboratory equipments you will use.
2. Never use broken or chipped glassware.
3. Make sure that all glassware's are clean before you using it.
4. Do not pick up broken glass with your bare hands. Use a pan and a brush.
5. If a Mercury thermometer breaks, do not touch the mercury.
6. Do not aim the mirror of your microscope directly at the sun. Direct sun light can damage the eyes.
7. Use care handling all sharp equipments, such as scalpels and dissecting needles.

Special precautions for working with live or preserved specimens

1. If live animals are used treat them gently. Follow instructions for their proper care.
2. Always wash your hands after working with live or preserved organisms.
3. Specimens for dissection should be properly mounted and supported. Do not try to cut a specimen while holding it in the air.
4. Do not open Petri dishes containing live cultures unless you are directed to do so.
5. Detergents (detol 5 – 10%) should be used to sterilize and clean benches, glassware and equipment.
6. Safety cabinet should be used while working with microbes.
7. Lab coats should be worn during the work in the lab.
8. Disposable items should be collected and autoclaved.

First Aid

1. **Injuries**: bleeding should be reduced using bandages; the wound should be cleaned with iodine alcohol mixture and wrapped with sterile bandage.
2. **Acid and fire burns**: body burns must be washed immediately with tap water. Eye burns must be washed using eye washer, special cream for burns can be used.
3. **Poisoning**: if any toxic chemical is swallowed, the mouth must be rinsed with water, in case of acid, milk is drunk, in case of alkaline, diluted acetic acid (vinegar) can be used.
4. Skin contamination requires washing with water and removal of contaminated clothing, if the contaminant is insoluble in water remove with soap and water.

PERMANENT SLIDE PREPARATION

A microscope slide is a thin flat piece of glass, typically 75 by 26 mm (3 by 1 inch) and about 1 mm thick, used to hold objects for examination under a microscope. Typically the object is placed or mounted on the slide, and then both are inserted together in the microscope for viewing. This arrangement allows several slide mounted objects to be quickly inserted and removed from the microscope, labeled, transported and stored in appropriate slide cases or folders. Microscope slides are often used together with a cover slip or cover glass, a smaller and thinner sheet of glass that is placed over the specimen. Slides are held in place on the microscope's stage by slide clips, slide clamps or a cross-table which is used to achieve precise, remote movement of the slide upon the microscope's stage.

The following points highlight the seven main processes involved in preparation of permanent slides.

The processes are:

1. Killing
2. Fixing and Hardening
3. Staining
4. Dehydration
5. Clearing
6. Mounting
7. Labelling.

Methods of Microscopic Preparations

For microscopic studies the specific material tissue organs or small organisms is mounted on a glass slide. There are two methods of mounting the material on slide.

- I. Temporary
- II. Permanent

I. Temporary mounting:-

The temporary mount is prepared either in glycerine, water or normal saline. The material is first washed in tap, then stained and differentiated. Drop of mounting medium (glycerine and water) is placed on centre of the slide. The material is then transferred into that drop. It is then covered neatly with a cover slip. The excess of glycerine or

water is absorbed by piece of blotting paper. Mount prepared by this method can be used for study only for few hours, after which material loses its original form due to diffusion and other post mortem changes.

II. Permanent mounting:-

But for the study of microorganisms, smaller animals and histological studies of tissues, an elaborate technique is employed for making their permanent preparations. These smaller objects are mounted in balsam on a slide. There is a series of processes by which a living organism or its tissue is made fit for microscopic examination in a permanent state. The utility of permanent preparation is that the animal cell or tissue remains as such without undergoing major changes.

The permanent preparation includes:

- (1) Killing and narcotisation.
- (2) Fixing.
- (3) Washing.
- (4) Staining.
- (5) De-staining or removal of excess of stain.
- (6) Clearing or de-alcoholization.
- (7) Mounting on slide.

1. Killing and Narcotisation :-

The first step in permanent preparation is killing instantaneously in order to prevent the change in form of the object as it has in living condition and immediately fixing the object. Sometimes killing is preceded by narcotisation. The narcotics used are chloroform, menthol, ether, alcohol, acetone, etc. the purpose of narcotisation and killing is important as to have the same form and chemically constructed tissue or organisms as it had during its lifetime. In certain cases, for smaller animals killing is heating done by the slide.

2. Fixing:-

Fixing is done with various fixative agents for histological elements. Fixative is essential in every type of microscopic preparations, either for sections or for whole mounts and also in larger specimens. The function of fixation is manifold:

- a. The tissues become hard and hardening resists further post-mortem changes.

- b. Fixative agent coagulates and renders insoluble elements of tissues which are dissolved in further processing.
- c. The fixative agent renders insoluble the various constituent elements of cells, alters their refractive indices and thus makes them optically differentiated under the microscope. Because of Brownian motion there is no possibility of material but we must bear in mind that fixed details are the coagulation artefact of the living structures.
- d. Various fixative agents generally used are absolute alcohol, 90% alcohol plus glycerine, picric acid, corrosive sublimate, formal, osmium tetra oxide and nitric acid with or without water. Depending upon the material, corrosive sublimate or alcohol Carnoy's fluid for cytological studies and other fixative for histochemical studies.

3. Washing :

Washing is essential as by this process the uncombined and excess of fixative agent is removed. The presence of fixative agent in tissues or cells will inhibit good staining. The washing agent depends upon the type of fixative agent used. As alcoholic picric acid in water is removed by 70% alcohol. Formal and corrosive sublimate are washed with water distillate. Sublimate is washed in alcohol.

4. Staining:-

The tissue or cell components are stained in various dyes. The dye makes the tissues distinct in its histological sphere. The various dyes are Orange G. Bordeaux red, Sudan's Congo red, Alizarins oxy quinoine, methylene blue, neutral red, borax carmine, haematoxylin, epicroindigo carmine, eosin and Gower's carmine.

Mainly two kinds of stains are used.

- I. **Nuclear stains:** Stains the nuclear parts of the cells, such as Delafield's or Ehrlich's haematoxylin.
- II. **Cytoplasmic stains** such as borax carmine, picro-indigo carmine, Gower's carmine and eosin, etc., which stain cytoplasm. For general staining borax carmine is used aqueous stains are prepared in water whereas alcoholic stains are prepared in alcohol. When a single stain is used the process is called as simple or single staining. In some cases two stains, i.e., nuclear and cytoplasmic are used and this is called as double

staining. Generally single stain is used for whole mounts but for protozoan's etc., both cytoplasmic and nuclear stains are used.

5. De staining:-

The removal of excess of stain is called as de staining or differentiation. De-staining agents are acid alcohol or acid water. The acid alcohol is used with alcoholic stains while acid water is used with aqueous stains.

6. De hydration :-

This process is meant for removal of water from the tissues. The dehydration prevents putrefaction or decaying and maintains the same shape and size of tissues or cells. The moisture or water in tissues absorbs various germs of destructive nature so that the tissue may be destroyed, hence the passing the mounting material through various grades of alcohol, such as 30, 50, 70, 90 and 100% alcohols. The tissue is soaked in gradually increasing strengths of alcohol. The lower grade prepared either from 90% or absolute alcohol. The dehydration is carried out in corked or glass-stopper tubes.

7. De-alcoholization or clearing:-

After dehydration, transparency in tissues is obtained by treating with a clearing agent, which removes alcohol and makes the tissue clear and transparent. The clearing agents are wood oil, clove oil, xylol and benzol, etc. Xylol is most commonly employed and it makes the tissues hard and brittle. Clove oil is a superior clearing agent especially in the whole mounts. It also possesses higher index of refraction than balsam mounting media.

8. Mounting:-

Mounting forms the end of permanent preparation the choice of mounting media is not much but they should have the same refractive index as that of the cleared tissue. The refractive index of such a stained, de hydrated and cleared cells is 1.54. Canada balsam or D.P.X has almost the same refractive index. Mounting is an easy process. The tissue is kept over glass slide in a drop of balsam and cover-slip is lowered slightly. After mounting, the excess of balsam on the slide, as generally happens with beginners, should be removed with cotton soaked with the balsam has dried. For much better finishing the edge of the cover-glass may be ringed with cement such as gold seal or a varnish. The air bubbles present in balsam under cover-glass should be removed by gentle heating.

During all the chemical bathing of tissues, two changes of each reagent are necessary. The time of keeping tissue in various reagents may vary from 5 to 15 minute.

9. Precautions and Instructions:-

1. The articles, such as slide, cover slips and instruments should be perfectly cleaned.
2. The working place should be kept in order.
3. During dehydration, the tissues should be kept in tightly closed cork or glass stopper tubes. The opened tube will spoil material by absorbing moisture from atmosphere. Even breathing closely with dehydrating tube is undesirable.
4. The change of solution is done very quickly, reducing time of exposure to atmosphere to minimum.
5. The chemicals used once should not be reutilized.
6. The Canada balsam used should be clean, dust-free and not viscous.

MAJOR PRACTICALS

COCKROACH

(*Periplaneta americana*)

Phylum : Arthropoda
Class : Insecta
Order : Blattodea
Family : Blattidae
Genus : *Periplaneta*
Species : *americana*

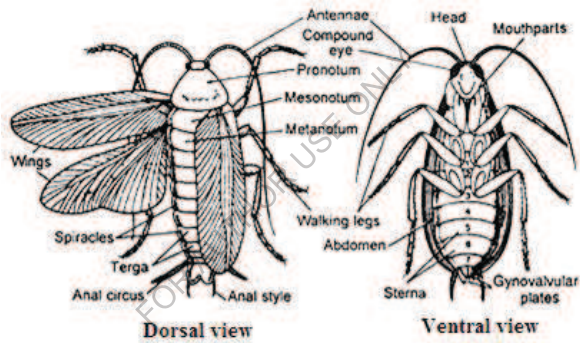


Fig.1. External morphology of Cockroach

COCKROACH – DIGESTIVE SYSTEM

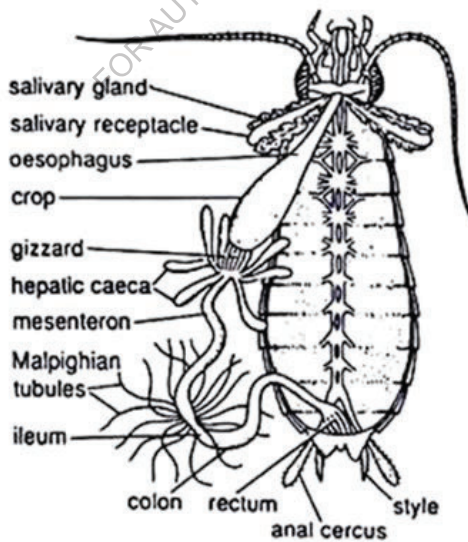
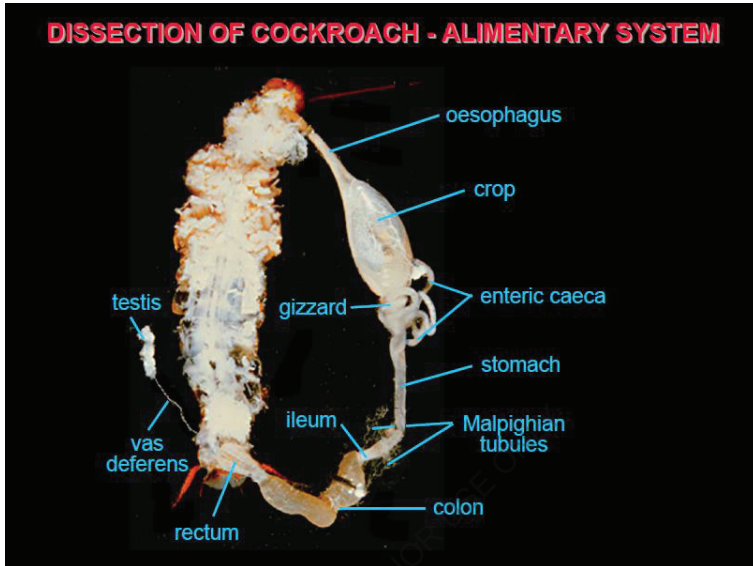


Fig.1aCocokroach-Digestive system

COCKROACH – NERVOUS SYSTEM

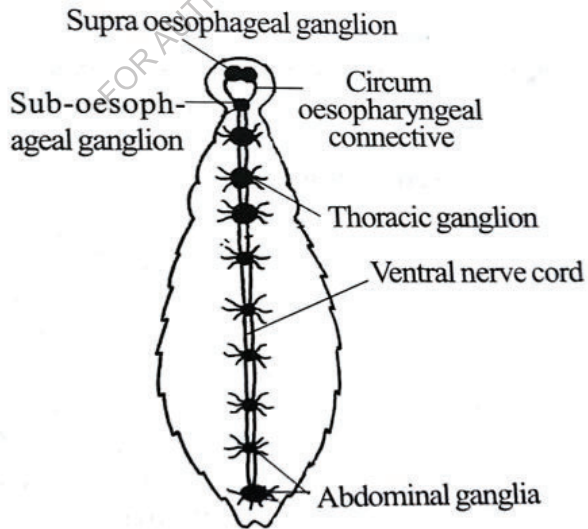
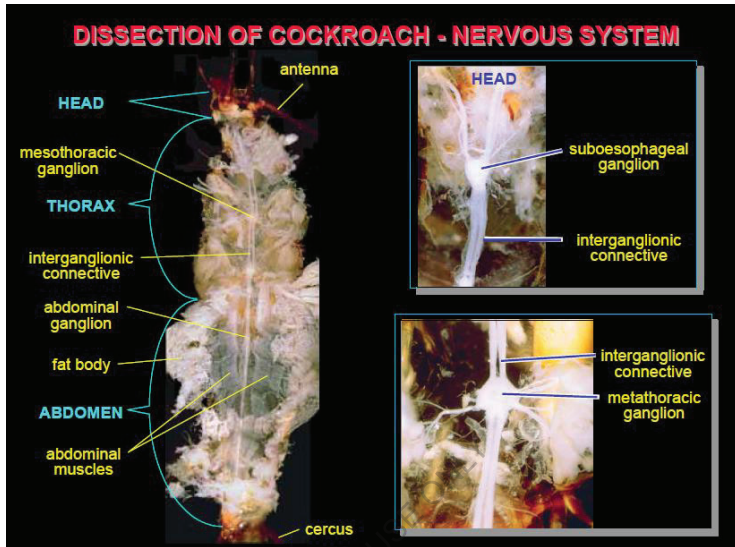


Fig.2. Cockroach-Nervous system

COCKROACH

MALE -REPRODUCTIVE SYSTEM

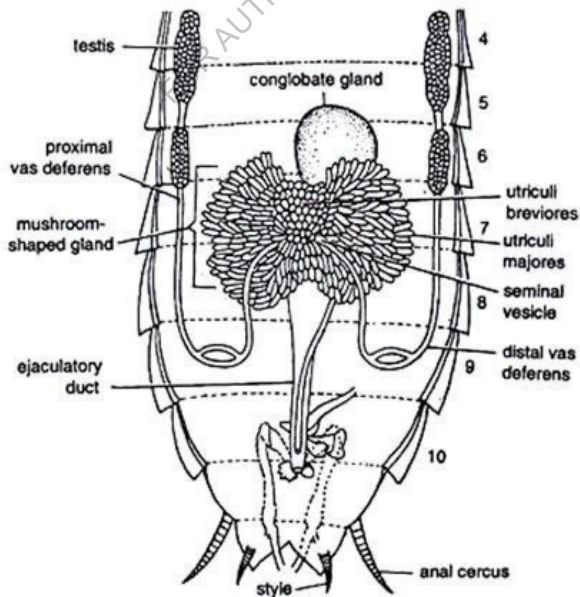
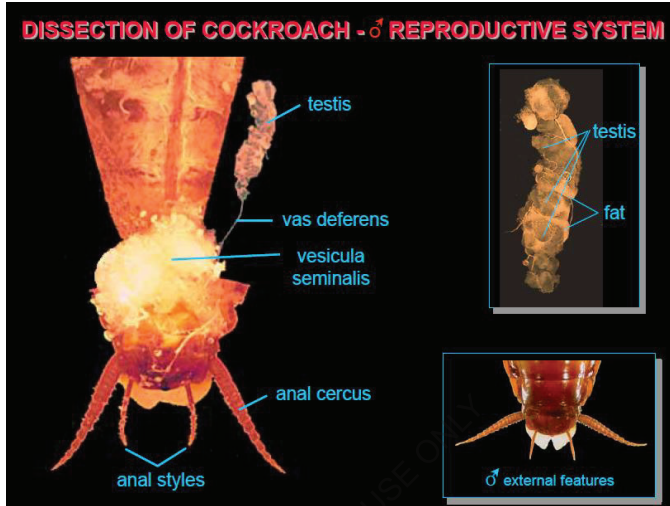


Fig. 3. Cockroach - male reproductive system

COCKROACH

FEMALE - REPRODUCTIVE SYSTEM

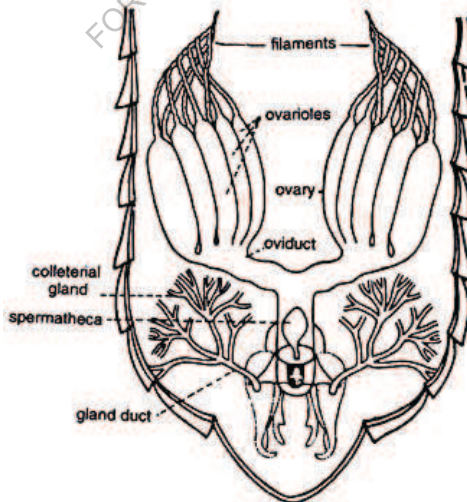
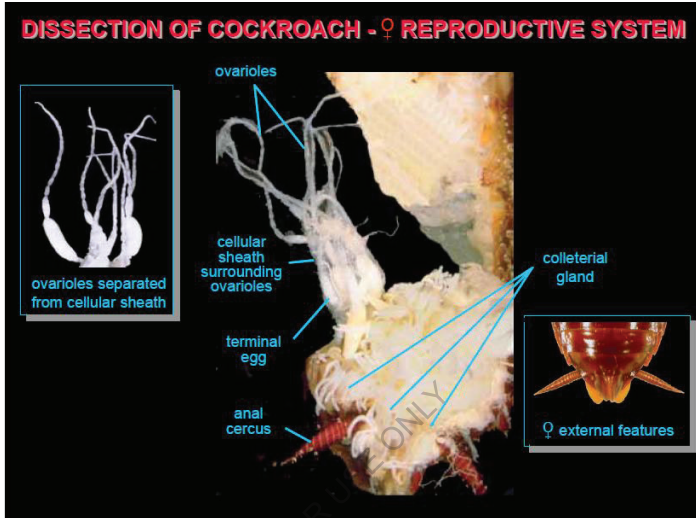


Fig.4. Cockroach-female reproductive system

MINOR PRACTICAL

EXTERNAL MORPHOLOGY OF HONEY BEE

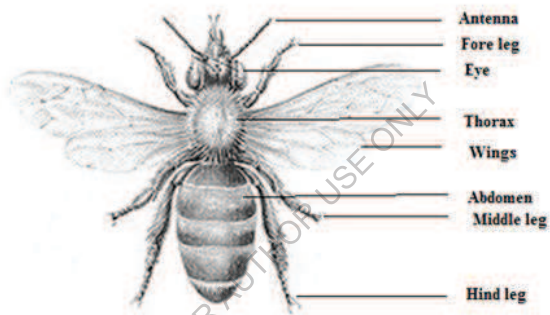


Fig 12. Honey bee

HONEY BEE – MOUTH PARTS

(CHEWING AND LAPPING TYPE)

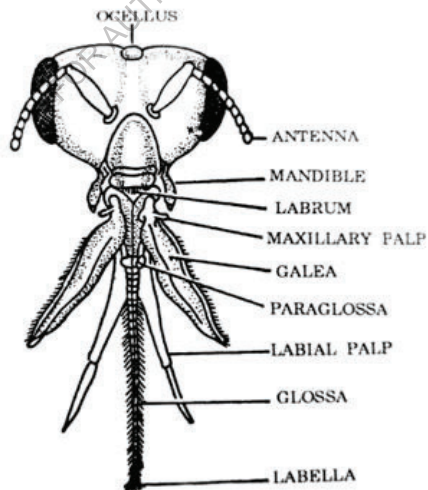
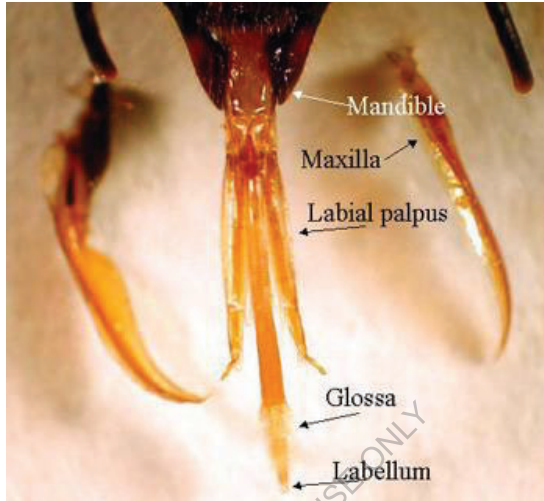


Fig.5. Mouth parts of honey-bee

EXTERNAL MORPHOLOGY OF MOSQUITO

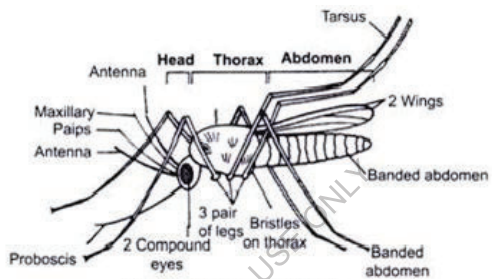


Fig. 16. Female Anopheles

MOUTH PARTS OF MOSQUITO (PIERCING AND SUCKING TYPE)

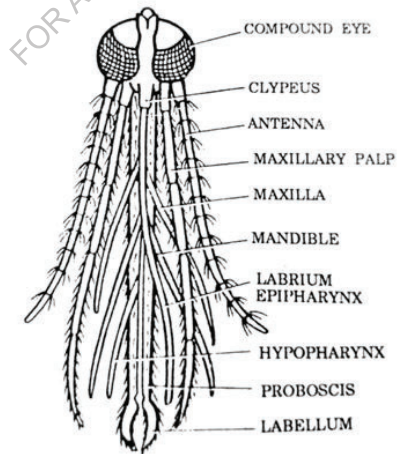
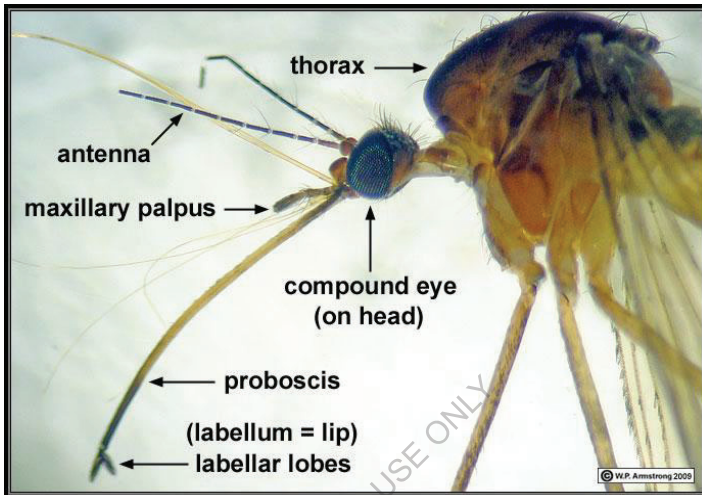


Fig.6. Mouth parts of female *Anopheles*

SPOTTERS

I. PROTOZOA

INTRODUCTION:

- Protozoans are **minute, unicellular (acellular)** organisms without any tissue grade of organisation.
- They are the **first animals** and hence the name protozoa.
- They are **free living or parasitic**.
- They are **asymmetrical or radially symmetrical or bilaterally symmetrical** animals.
- They have **protoplasmic** grade of organisation.
- All the activities carried out by the **cytoplasm** of the single cell.
- **Tissue and organs** are absent in protozoans.
- Locomotion is by **flagella, cilia or pseudopodia**.
- Nutrition is **holophytic, holozoic, saprozoic or parasitic**.
- Digestion is **intracellular**.
- Respiration and excretion occurs by **diffusion**.
- In freshwater protozoans, osmoregulation is carried out by **contractile vacuoles**.
- **Encystment** is a common phenomenon.
- Reproduction occurs by **asexual or sexual methods**.

ENTAMOEBA HISTOLYTICA

SYSTEMATIC POSITION

Phylum : Protozoa
Sub Phylum : Sarcomastigophore
Class : Rhizopoda
Order : Lobosa
Genus : *Entamoeba*
Species : *histolytica*

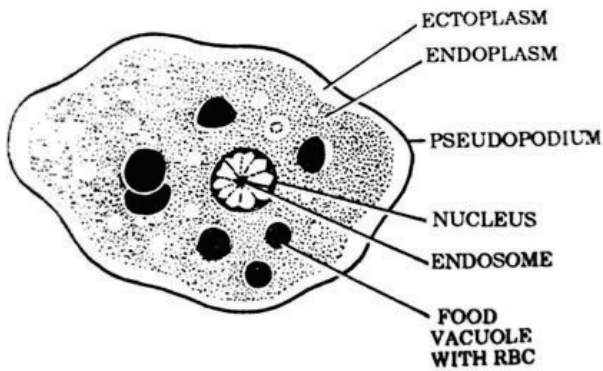


Fig.7. *Entamoeba histolytica*

ENTAMOEBA HISTOLYTICA

Distribution:-

Entamoeba histolytica is worldwide on distribution, but it's prevalence is greater in tropics and subtropics than in the temperate zones. *E. histolytica* infecting humans and other primates causing **amoebiasis** and it infect about 50 million people worldwide.

Habits and Habitat:-

1. *Entamoeba histolytica* is a microscopic endoparasite of man. It is commonly found in the upper part of the large intestine and is very often lodged in the liver, lungs, brain and testes.
2. The size varies from 0.06 μm to 10 μm in diameter. It is dimorphic occurring in two forms: (i) **Trophic or trophozoite**, (ii) **Precystic or minuta**.
3. The active (trophozoite) stage exists only in the host and in fresh loose feces; cysts survive outside the host in water, soil and on foods, especially under moist conditions on the latter.
4. Some risk factors for amoebiasis (The active trophozoite stage) include consuming contaminated food or water, association with food handlers whose hands are contaminated, contact with contaminated medical devices such as colonic irrigation devices and being pregnant.
5. When cysts are swallowed they cause infections by excysting (releasing the trophozoite stage) in the digestive tract.
6. It occurs in the colon of man and feeds on the mucous membrane destroying the tissues by an enzyme which it secretes. The *E. histolytica* can actually 'bore' into the intestinal wall, causing lesions and intestinal symptoms and it may reach the blood stream. From there, it can reach different vital organs of the human body, usually the liver, but sometimes the lungs, brain, spleen, etc. Ingested red blood cells are sometimes seen in the amoeba cell cytoplasm.
7. Symptoms can include fulminating **dysentery**, **bloody diarrhea**, **weight loss**, **fatigue**, **abdominal pain** and **amoeboma**.
8. Generally **Paromomycin** (Humatin), **Iodoquinol** (Yodoxin) and **Diloxanide furoate** are the antibiotics available to treat *E. histolytica*.

PARAMECIUM CAUDATUM

SYSTEMATIC POSITION

Phylum	: Ciliophora
Class	: Oligohymenophorea
Order	: Peniculida
Family	: Parameciidae
Genus	: <i>Paramecium</i>
Species	: <i>P. caudatum</i>

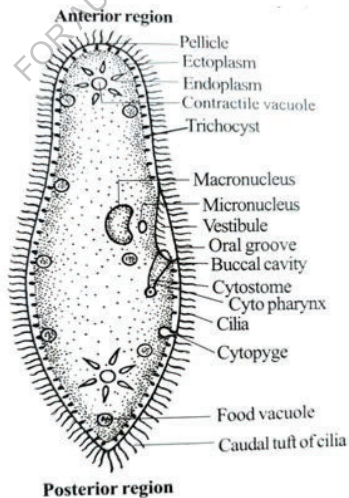


Fig.8. *Paramecium caudatum*

PARAMECIUM CAUDATUM

COMMENTS:

1. *P. cadatum* is a **microscopic, unicellular protozoan**. Its size ranges from 170 to 290um or up to 300 to 350um.
2. *Paramecium caudatum* is a free-swimming ciliate found abundantly in pools, lakes, ditches, ponds, freshwater and slow flowing water that is rich in decaying organic matter.
3. It has **cosmopolitan** distribution.
4. Elongated body (0.25 mm) with rounded anterior and bluntly-pointed posterior end and its looks like a slipper. Hence it is called as **slipper animalcule**.
5. The entire surface is covered by vibratile cilia. This arrangement is called **holotrichous**. The posterior end of the body forming a caudal tuft of cilia, thus named caudatum.
6. A wide depression extends obliquely backwards towards the right and ends behind the middle of the body. **Peristome** leads into the gullet or **cytopharynx**.
7. Its whole body is covered with a flexible, thin and firm membrane called pellicles. These pellicles are elastic in nature which supports the cell membrane. It is made up of a gelatinous substance.
8. *Paramecium* consists of two types of vacuoles: **contractile vacuole** and **food vacuole**.
9. Protoplasm has an outer clear **ectoplasm** and inner dense **endoplasm**.

The ectoplasm contains.

- (a) basal granules from which arise the protoplasmic fibril which runs in the axis of the cilium and
- (b) Trichocysts lying a little deeper in the cortex. Trichocysts are filled with a dense refractive fluid containing swelled substances. There is a conical head on the spike at the outer end. Trichocysts are perpendicular to the ectoplasm.

The endoplasm contains

- (a) a large elongated macro nucleus,
 - (b) a small micro-nucleus
 - (c) Several food vacuoles and two contractile vacuoles. Contractile vacuoles (Osmoregulation) contract alternately.
10. Paramecium reproduces asexually through **binary fission**. The micronuclei during reproduction undergo mitosis while the macronuclei divide through amitosis. Each new cell, in the end, contains a copy of macronuclei and micronuclei after the cell undergoes a transverse division. Reproduction through binary fission may occur spontaneously.

II. COELENTERATA

INTRODUCTION

- Coelenterate is the Multi cellular organisms with tissue grade of organization and are aquatic, mostly marine except a few freshwater forms like Hydra.
- They are sedentary or free swimming, solitary or colonial and show the radial or biracial with central gastro vascular cavity opening to exterior by mouth.
- Diploblastic body consisting of outer layer of ectoderm, inner endoderm and an intermediate non cellular gelatinous mesoglea.
- Acoelomate.i.e. coelom is absent. Mouth is encircled by tentacles bearing nematocysts (stinging cells) meant food, adhesion, offence and defence.

AURELIA AURITA

SYSTEMATIC POSITION:

Kingdom : Animalia
Phylum : Cnidaria
Class : Scyphozoa
Order : Semaeostomeae
Family : Ulmaridae
Genus : *Aurelia*
Species : *A. aurita*

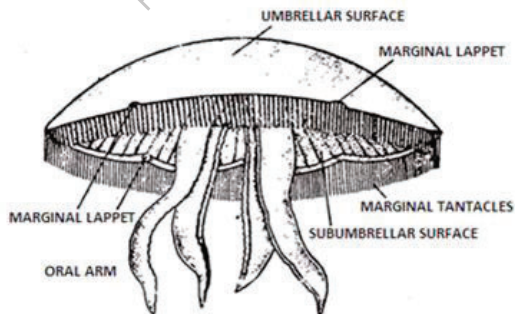


Fig.9. *Aurelia aurita*

AURELIA AURITA

Comments:

1. *Aurelia* is commonly known as **Jelly-fish**. It is a solitary, marine, medusoid form inhabiting warm and temperate seas throughout the world.
2. Body is transparent, bluish white in colour. Inner organs are visible from body surface.
3. Circular body measuring about 90 mm diameter having convex aboral (ex umbrella) and concave oral (sub umbrella) surface.
4. All the four comers of the mouth are drawn out into four long, mobile, delicate, frilled and tapering processes called the **oral arms**. The oral arms are provided with a ciliated groove on its inner side and they are also provided with numerous **nematocysts**.
5. Four horse-shoe-shaped pinkish gonads, which later become circular, are situated between oral arms in the centre of the subumbrellar surface.
6. In the centre of subumbrellar surface, there is a well developed but short and inconspicuous manubrium bearing a large squarish mouth.
7. Sub umbrellar margin contain marginal tentacles having stinging cells and 8 marginal lappets having sense organs called **tentaculocysts** (rhopalium)
8. Jelly fish is **carnivorous** feeding on small animals like molluscs, crustaceans, protozoan's, nematodes etc.,
9. It swims by rhythmic contractions of umbrella surface.
10. There are no specialised organs for respiration and excretion. These processes are presumably performed by the whole body surface.
11. The nerve cells and their fibres form two nerve nets joined to the sensory cells and muscle fibres.
12. *Aurelia* is dioeciously i.e. male and female individuals are separate. The four gonads (testes or ovaries) lie inter- radially on the floor of the gastric pouches.
13. The gonads are horse-shoe-shaped and reddish in colour. Gonads of *Aurelia* are endodermal in origin unlike those of *Obelia* medusae which are ectodermal in origin. Sex cells originate in the endoderm of gonads. The ripe gametes are shed into the stomach and passed out through the mouth.
14. The developmental stages of *Aurelia* with the larval stages namely **Planula larva and Ephyra larva**.
15. Life cycle shows alteration of generation.

III. PLATYHELMINTHES

INTRODUCTION:

- Platyhelminthes are bilaterally symmetrical and dorsoventrally flattened.
- Body wall is Triploblastic (form 3 germ layer) epidermis, mesoderm, and endodermis) below the endodermis longitudinal.
- Circular and oblique muscles are present. Acoelomate.i.e (true coelom is absent) and un segmented except class cestoda.
- They are free living, parasite or commensals. Adhesive organ like hooks, suckers and spine are present with adhesive secretion.

TAENIA SOLIUM

SYSTEMATIC POSITION:

Phylum	:Platyhelminthes
Class	:Cestoda
Subclass	:Eucestoda
Order	:Taenioidea
Genus	: <i>Taenia</i>
Species	: <i>solium</i>

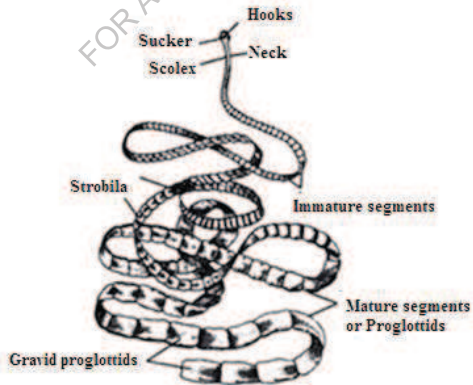


Fig. 10. *Taenia solium*

TAENIA SOLIUM

Comments:

1. *Taenia solium*, the **pork tapeworm** of man, as adult lives in the intestine of man leading an **endoparasitic** life. Its life history is completed in two hosts, i.e., **digenetic**; man being the primary host and pig as secondary host.
2. The body of *Taenia solium* is long, dorsoventrally flattened, narrow, ribbon-like, reaching a length of two to three metres. The colour of the body is opaque-white. Body consists of scolex, neck and strobila or body segments.
3. The anterior end of the body of *Taenia* has a knob-like scolex. The scolex is smaller than the head of a pin about 1 mm in diameter.
4. In addition to suckers, scolex contain rostellum, which is pigmented and provided with a double row of hooks (number of hooks are about 28 to 32).
5. Body comprises of large number of segments nearly 800 or more, these segments called **proglottids**.
6. Each segment (Proglottids) contain one set of male and female reproductive organs, part of excretory, nervous system and a lateral genital opening.
7. The alimentary canal is altogether absent in *Taenia solium*, therefore, its mode of nutrition is **saprophytic**, i.e., it absorbs digested liquid food from the intestine of the host through its general body surface.
8. The mode of respiration is anaerobic, as the tapeworm lives in the intestinal contents which is oxygen free. The energy is derived from the breakdown of glycogen. During the breakdown process, carbon dioxide, fatty acids and other organic acids are produced as waste products.
9. Life cycle involves two hosts
 - (i) Man-definitive host
 - (ii) Pig-intermediate host
10. When pigs eat contaminated human faeces becomes infected by bladder worm (*Cysticercus* larva).
11. Human are infected by eating pork (pig meat) and *Taenia solium* develops into intestine of adult man.
12. The tapeworm exhibits a number of adaptive features to live comfortably in the intestine of the human beings.

***TAENIA SOLIUM*-SCOLEX**

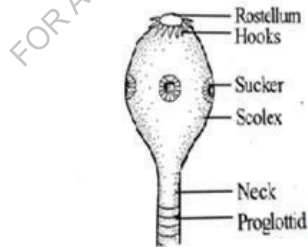


Fig.10.a. Scolex of *Taenia solium*

***TAENIA SOLIUM*-SCOLEX**

COMMENTS:

1. The body of *Taenia solium* is long, dorsoventrally flattened, narrow, ribbon-like, reaching a length of two to three metres. The colour of the body is opaque-white. Body consists of **scolex**, **neck** and **strobila** or body segments.
2. The anterior end of the body of *Taenia* has a knob-like scolex.
3. The scolex is smaller than the head of a pin about 1 mm in diameter with 4 cup-like muscular suckers having radial muscles, and an anterior round prominence, the **rostellum** having about 22 to 32 curved, chitinous hooks in two circles, the inner circle with larger hooks and outer circle with smaller ones.
4. The long and short hooks alternate with each other. Each hook is made of a base by which it fixes, a handle directed towards the apex and a conical blade directed outwardly. The rostellum can be protruded slightly.
5. The scolex with its suckers and hooks is an organ of attachment to the intestinal wall of the host, thus, working as an organ of adhesion or the holdfast.
6. The scolex is, sometimes, wrongly referred to as '**head**' but it cannot be the head because it is neither related to food- perception nor an organ of catching the food.

T.S OF *TAENIA SOLIUM*

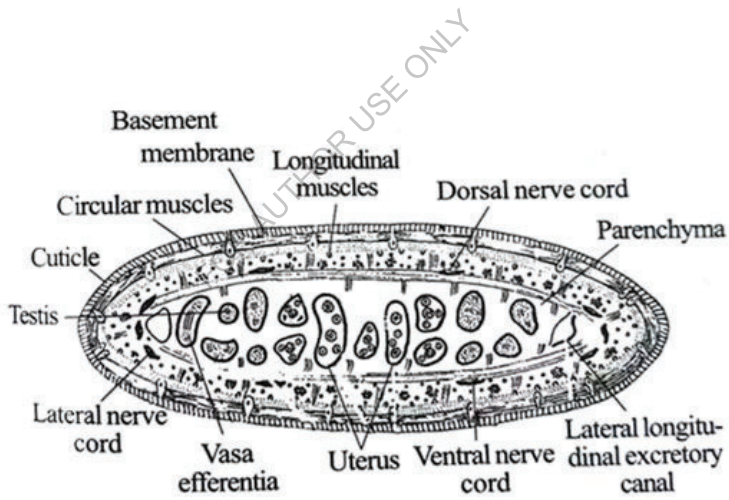


Fig. 11. T.S OF *TAENIA SOLIUM*

LIVER FLUKE

(*FASCIOLA HEPATICA*)

Phylum: Platyhelminthes

Class: Rhabditophora

Order: Plagiorchiida

Family: Fasciolidae

Genus: *Fasciola*

Species: *F. hepatica*

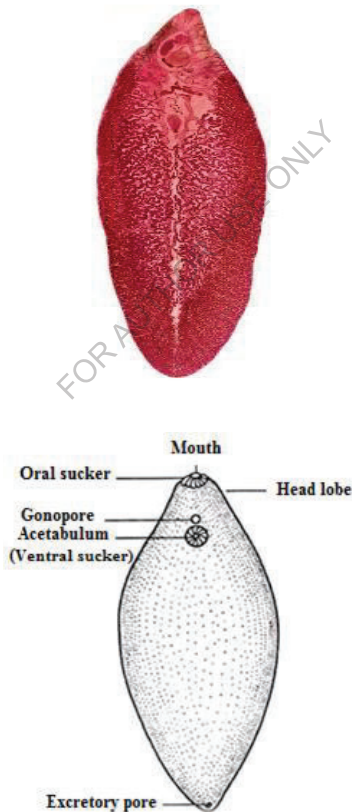


Fig13 *Fasciola hepatica*

LIVER FLUKE

(*FASCIOLA HEPATICA*)

1. Liver Fluke is a soft bodied, flattened leaf-like animal (Fig. 14.10) and exhibits bi-lateral symmetry. The size varies from about 1.0 to 2.5 cm in length but the width does not generally exceed 1.0 cm.
2. It is an **endoparasite**. It is commonly known as **liver fluke**.
3. It is a flattened worm. Hence it is included in the phylum Platyhelminthes and class Trematoda.
4. It lives inside the bile duct of sheep, goats and cattle. It causes diseases called **liver rot** in sheep.
5. It is narrow at the anterior end, broad at the middle and tapers towards the posterior end.
6. At the anterior end there is a triangular projection, the **head lobe**.
7. It has two suckers, an oral sucker or anterior sucker at the tip of the head lobe and a ventral sucker or **acetabulum** behind the head lobe.
8. The suckers help in the attachment of the parasite to the host.
9. Between the two suckers there is a **genital opening or gonopore** is present.
10. The excretory pore is present at the hind end of the animal.
11. Liver Fluke reproduces sexually and the same individual bears both the sexes (hermaphrodite).
12. The life cycle of liver fluke is completed through five larval stages, such as **Miracidium larva, Sporocyst larva, Redia, Cercaria and Metacercaria**.
13. Liver Fluke is a **digenetic** trematode parasite because it completes its life cycle through two hosts. The definitive (or called primary) host is sheep or cattle than the man although several hundred cases of human **fascioliasis** have been reported from Latin America. The intermediate (= secondary) host is a gastropod snail known mainly *Lymnaea truncatula*.
14. In sheep and cattle, oral administration of hexachloroethane and carbon tetrachloride are useful to kill the parasites. Human **fascioliasis** are treated by intramuscular injections of emetine hydrochloride for 10 days.
15. To eradicate the intermediate hosts, copper sulphate and pentachlorophenate can be used in the ponds, streams or in other snail's habitats, which may kill the snail hosts.

C.S OF *FASCIOLA HEPATICA*

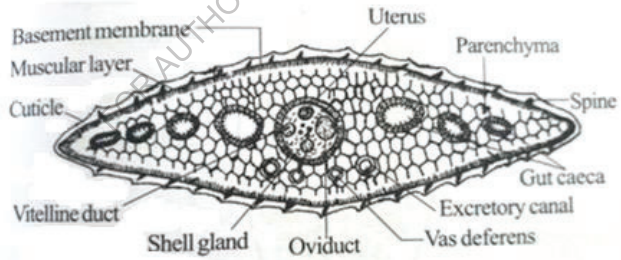


Fig.14. C.S OF *FASCIOLA HEPATICA*

ASCHELMINTHES

INTRODUCTION:-

- Phylum Aschelminthes (Round worms) inhabit aquatic and terrestrial environment and are well known parasites.
- They are bilaterally symmetrical and unsegment worms.
- Triploblastic and pseudo coelomate animals with organ system grade of body organization.
- Body size mostly small, some microscopic, while others few millimetres to a meters or even more in length. Body mostly cylindrical or flattened, body wall with cuticle and cilia absent.
- The digestive includes straight and complete alimentary canal with anus, muscular and highly specialized pharynx.

ASCARIS LUMBRICOIDES

SYSTEMATIC POSITION:

Phylum : Aschelminthes
Class : Nematoda
Order : Ascaridida
Family : Ascarididae
Genus : *Ascaris*
Species : *lumbricoides*

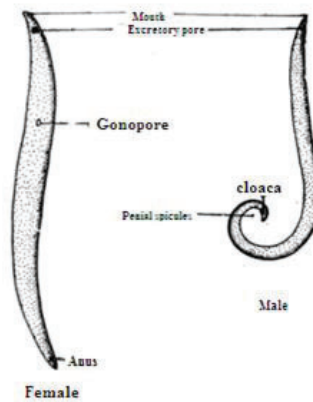


Fig.12. *Ascaris lumbricoides*

ASCARIS LUMBRICOIDES

COMMENTS:

1. A **roundworm** called *Ascaris lumbricoides* infects a person's small intestine and acts as an **endoparasite**, getting nutrients from its host's intestinal tract.
2. It is cosmopolitan in distribution.
3. It shows **sexual dimorphism** with separate male and female individuals. Males measure 15-30 cm in length and female 20-35 cm.
4. The tail end of male is ventrally curved containing **cloacae aperture**, through which two spicules project.
5. The tail end of the female is straight. Anterior and end exhibits same structures in both male and female.
6. Mouth is situated at anterior end. It is guarded by three lips.
7. Anaerobic respiration is noticed.
8. Excretory pore lies at the distance of 2 mm. From anterior end. No intermediate host is required.
9. Life cycle includes a rhabdiform larva and an extra intestinal migration.
10. **Prevention:**
 - People should also wash any utensils and cooking surfaces after each use.
 - Filter and boil water before drinking it.
 - Wash hands with soap and hot water before handling food and eating.
 - Avoid common bathing areas, especially if they appear unclean.
 - Peel and cook any vegetables or fruit grown in unsanitary conditions.
 - Ensure children clean their hands with soap and water after playing outside.
11. **Treatment :** The worm will treat most cases of **ascariasis** with antiparasitic drugs like Albendazole (Albenza), Ivermectin (Stromectol) and Mebendazole.

VIII. ECHINODERMATA

INTRODUCTION:-

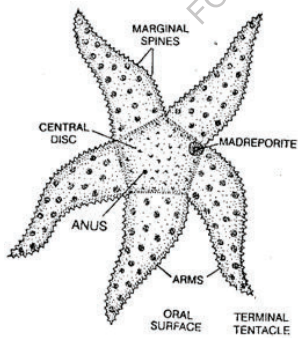
- Phylum Echinodermata includes exclusively marine invertebrates displaying pentamerous radial symmetry and an endoskeleton of calcareous plates and spines. Jacob Klein gave the name Echinodermata.
- This phylum is a collection of about 7,000 living species and constitute some of the most beautiful members of sea fauna, such as starfishes or sea stars, sea urchins, sea cucumber, sea lilies and sand dollars etc.,
- The name 'starfish' is, however, misleading as these animals are not true fishes. A more suitable name suggested for them is 'sea star'. Common genera of sea stars are *Asterias*, *Pentaceros*, *Astropecten*, *Solaster*, *Heliaster*, *Luidia*, *Asterina* etc., Genus *Asterias* includes about 150 species of which most common is *A. Rubens*.

STAR FISH

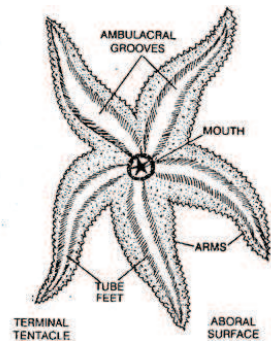
(*ASTERIUS RUBENS*)

SYSTEMATIC POSITION:

Phylum : Echinodermata
Class : Asteroidea
Order : Forcipulatida
Family : Asteriidae
Genus : *Asterias*
Species : *rubens*



Dorsal view



Ventral view

Fig. 16. Star fish (*Asterias rubens*)

STAR FISH

Habit and habitat:-

Asterias are free-living marine animals that can be found at all water depths as well as crawling over rocks and shells. Asterias is a marine form and found in the sandy and rocky parts of the sea.

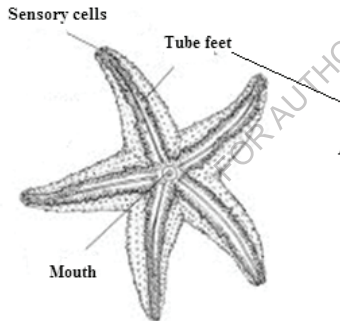
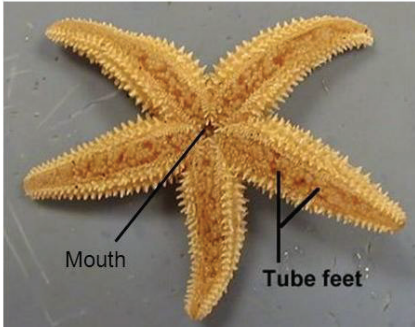
Distribution:-

Asterias is found in shallow water in North Temperate seas and found abundantly on North-Atlantic coast. It is found in abundance in India and U.S.A.

COMMENTS:

1. Asterias is commonly known as starfish or sea star.
2. Average size of Asterias ranges from 10 to 25 cm in diameter. The body has a central disc and five arms.
3. It is **radially** symmetrical animal.
4. The body is star shaped, consisting of a central disc with five radiating arms which are broad at their base and tapering towards their extremities.
5. The body consists of two surfaces, namely aboral surface and oral surface.
6. The oral surface bears the mouth and five ambulacral grooves with water vascular system. The tube feet occur in each ambulacral groove.
7. Aboral surface bears a number of short, stout spines arranged in irregular rows, numerous dermal branchiae among spines and a smaller aperture, the anus which is situated near the centre.
8. Pedicellariae are very small, microscopic bodies scattered all over the body.
9. Sexes are separate. Fertilization is external.
10. Development includes a free swimming bipinnaria larva.

STAR FISH-TUBE FEET



Basic structure of tube feet

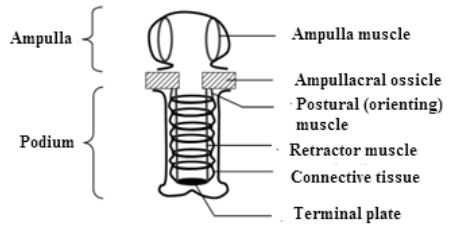


Fig.16 (a). Star fish-Tube feet

STAR FISH-TUBE FEET

COMMENTS:

1. **Tube feet** are small active tubular projections on the oral face of an echinoderm, whether the arms of a starfish or the undersides of sea urchins, sand dollars and sea cucumbers . They are part of the water vascular system.
2. Tube feet function in **locomotion**, **feeding** and **respiration**. The tube feet in a starfish are arranged in grooves along the arms. They operate through **hydraulic pressure**.
3. They are used to pass the food to the mouth at the center and can attach to surfaces. A starfish that is overturned simply turns one arm over and attaches it to a solid surface and levers itself the right way up.
4. Tube feet allow these different types of animals to stick to the ocean floor and move slowly. Tube feet consist of two parts: **ampullae** and **podia**.
5. Ampullae contain both circular muscles and longitudinal muscle, whereas the podia contain the latter only. The podia use a particular glue (no suction) to attach to the substratum.
6. Contraction of the ampulla pump by ampulla muscles increases the pressure in the ampulla cavity, which forces its fluid into the fluid-filled cavity of the podium. The resultant increase in pressure within the podium causes tube-foot extension.
7. The terminal sucker of the podium enables the podium to attach to the substrate and hence to apply force to the substrate.
8. Coordinated action of the tube feet brings about slow creeping locomotion. One arm temporarily dominates and leads the way, according to which arm receives the strongest positive stimulus (induced arm dominance).
9. In many starfish, small animals, even fish, captured by the pedicellaria are passed to the podia which convey them to the mouth. In predatory starfish, the podia may be used to help gain access to food, for example by prizing the valves of muscles and oysters apart. A single podium may exert a force of 25-30 grams and working together they may exert a pressure of nine pounds per square inch.
10. In burrowing starfish, such as *Astropecten*, the tube-feet lack suckers and are used for both walking and burrowing into sand.

SEA ANAEMONE HERMIT CRAB

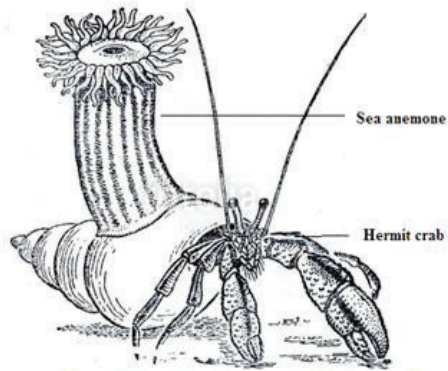


Fig. 17. Sea Anaemone & Hermit crab

SEA ANAEMONE & HERMIT CRAB

Commensalism

The association between two different species of organisms where one is benefited and another one neither benefited nor harmed is “**commensalism**”.

Hermit crabs and sea anemones have a symbiotic relationship, meaning they live in harmony together. It’s a stable connection that sidesteps the typical prey and predator relationship found in nature. The type of symbiosis they engage in is called commensalism. Commensalism means one organism benefits and the other organism isn’t harmed by the partnership. Neither the hermit crab nor the sea anemone is negatively affected by their symbiotic and commensalistic relationship.

Protection

The hermit crab gains protection from predators by its relationship with the sea anemone. Scientists aren’t certain if the hermit crab uses a mechanical or chemical means to communicate a potential threat to the anemone, but the anemone responds to the alert just the same. The sea anemone spreads out long stinging threads over the hermit crab like a bright pink curtain. The anemone also extends its stinging tentacles out as additional protection. A hermit crab is less likely to be eaten by a larger predator fish if he has an anemone onboard.

Food

The sea anemone will eat just about anything in the sea, it gets to eat whatever tidbits the hermit crab leaves behind. The hermit crab does the work of capturing prey and the sea anemone cleans up the leftovers. It’s a steady food supply for the sea anemone.

Mobility

Sea anemones move very little, if at all. An added benefit of hitching a ride on a hermit crab’s shell is getting exposed to a much larger area to forage for food. While the hermit crab is trolling around on the sea floor, the anemone can make use of the time traveling on his mobile home to trap and collect plankton and small fish.

EPHYRA LARVA OF *AURELIA*

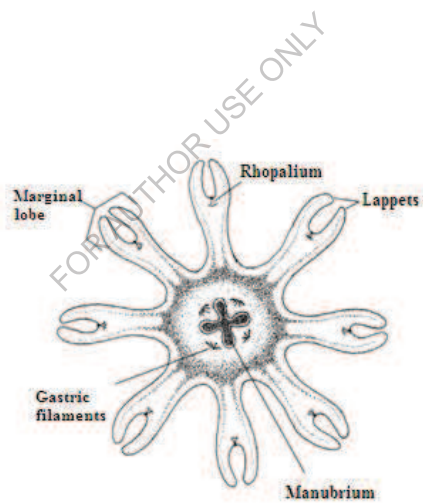
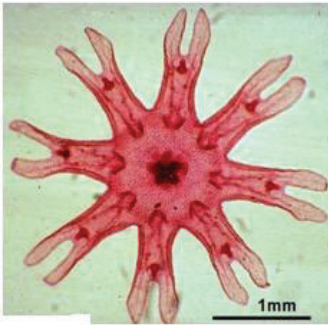


Fig.15. Ephyra larva of *Aurelia*

EPHYRA LARVA OF *AURELIA*

COMMENTS:

1. The life cycle of free-swimming scyphozoan jellyfish typically consists of three stages.
2. A sessile polyp (scyphistoma) stage asexually buds off young medusae from its upper end, with each such ephyra growing into an adult.
3. The ephyra is a young medusoid form. It has eight notched lobes or arms which are per-radial and inter-radial. Each lobe has two exaggerated marginal lappets with a deep groove between them having a short tentacle which becomes a tentaculocyst.
4. An ephyra bears a short stomach or gastric cavity with a short manubrium and four-sided mouth on its subumbrellar surface. The gastric cavity grows into the lobes to form branched per-radial and inter-radial canals, later the ad-radial canals also appear. Four pairs of gastric filaments form in the inter-radial mesenteries.
5. The ephyra feeds largely on protozoans which are caught by the lappets and transferred to the mouth.
6. The ephyra grows in size, the spaces between notched lobes fill up, mesogloea increases tremendously pushing the endoderm layers together to form an endoderm lamella except in the enteron, four oral arms and marginal tentacles appear, the septal funnels become subgenital pits, and by summer the ephyra becomes an adult medusa.
7. Thus, the stages in the life history of *Aurelia* are male and female medusae → gametes → zygotes → ciliated planula larvae → hydratubae → scyphistomae → ephyra → medusa.
8. In *Aurelia Aurita*, there are several phases that show variations which are found in the life-cycle of other Scyphozoa. Very large eggs of *Aurelia* become actinula larvae (an advanced type of planula which is like a polyp and has a short stem) which directly form ephyrae larvae. Smaller eggs form planula larvae which become hydratubae and then scyphistomae which strobilate ephyrae.
 9. Strobilation is brought about by intensive feeding followed by a drop in the temperature. In *Aurelia* if the food is plentiful and the temperature is low, then many ephyrae are produced at one time (polydisc strobilation). If there is less food and the temperature is high then ephyrae are produced one at a time (monodisc strobilation).

REDIA LARVA OF LIVER FLUKE

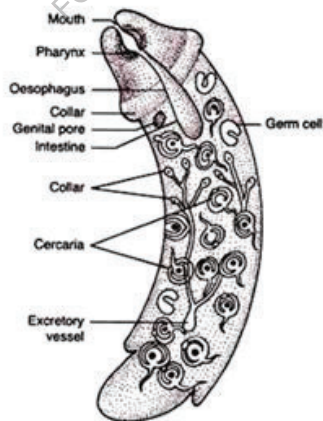


Fig.18 .Redia larva

REDIA LARVA OF LIVER FLUKE

Liver fluke is digenic parasite. It completes its life cycle in two hosts, namely sheep and snail called *Limnaea trunculata*. Sheep is the primary host and snail is the secondary host. Its development is indirect since there are larval stages.

COMMENTS:

1. Redia larva develops from the germ cells of **sporocyst**. The redia larva has the following salient features.
2. It lives in the digestive glands of the **snail**.
3. It is **cylindrical** in shape.
4. The body is covered by **cuticle**. The anterior end has a mouth.
5. Behind the mouth, there is a muscular ring called **collar**.
6. Behind the collar there is an opening called **birth pore**.
7. Near the posterior end a pair of projection is called **lappet**. It is used for locomotion.
8. The mouth leads into a pharynx which ends in a sac-like intestine. The pharynx is surrounded by a group of pharyngeal glands.
9. Two **protonephridia** are located inside the body. Each protonephridium is formed of many flame cells. It opens to the outside by **nephridiopore**.
10. The cavity of larva filled with **germ cells**.
11. The germ cells of redia develop into daughter redia.
12. The germcells of daughter redia develop into the next larva called **cercaria**. They come out through birth pore.

CERCARIA LARVA OF LIVER FLUKE

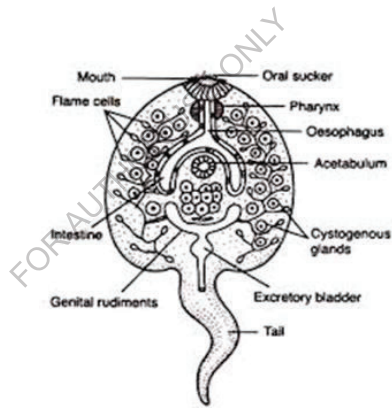


Fig. 19. Cercaria larva

CERCARIA LARVA OF LIVER FLUKE

Liver fluke is **digenic** parasite. It completes its life cycle it's in two hosts, namely sheep and snail called *Limnaea trunculata*. Sheep is the primary host and snail is the secondary host. Its development is indirect since there are larval stages.

COMMENTS:

1. The Cercaria larva is fourth developmental stage of **liver fluke**.
2. The Cercaria develops from the germ cells of redia larva. Each redia produces about 20 cercaria. The cercaria has the following salient features:
3. It is a **free swimming** larva.
4. It is a **tadpole** shaped.
5. It has an oval body and a tail.
6. The body is covered by **cuticle**.
7. It has **two suckers**, an **oral sucker** and the **acetabulum**.
8. It has a simple alimentary canal. It is formed of mouth, pharynx, oesophagus and 'U' shaped intestine.
9. Numerous flame cells are located inside the body. The flame cells of each side are connected together by an excretory vesicle. From the vesicle an excretory duct arises. It runs into the tail and bifurcates. These bifurcations opens out by nephridiopores.
10. The body wall contains many cytogenous gland.
11. The body cavity fills with group of germ cells.
12. The cercaria lives for three days and it is transformed into another larva called **metacercaria**.

NAUPLIUS LARVA

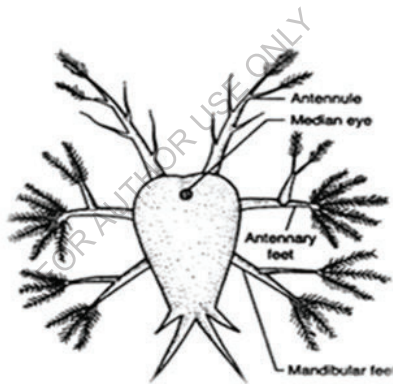


Fig. 20. *Nauplius* larva

NAUPLIUS LARVA

COMMENTS:

1. **Nauplius larva** the first, free-swimming, planktonic larva of most marine and some freshwater crustaceans.
2. It is a free swimming larva and more or less triangular in shape.
3. There is a single, median eye at the front of the head.
4. Mandible along with antennae is helpful to food collections.
5. There are only three pairs of appendages, the first and second antennae, and the mandibles. The second antennae and mandibles bear swimming setae.
6. Additional trunk segments and appendages appear with successive moults, the increments proceeding from anterior to posterior.
7. The late Nauplius stages are often called metanauplii.
8. In some forms nauplius larva develops straight away into adult, but in many other crustaceans forms it give rise to other intermediate larval forms like metanauplius, protozoaea, zoea, crypsis, mysis, megalopa, phyllosoma and alima.
9. The mouth and alimentary canal are absent in Nauplius larva and it contains numerous germ cells or primitive ova.

MYSIS LARVA OF *PENAEUS*

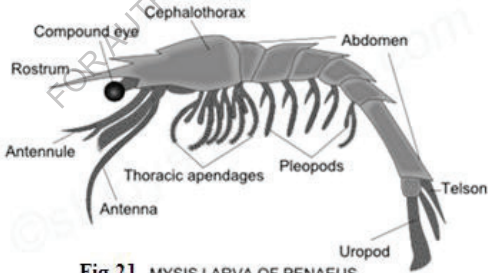


Fig.21. MYSIS LARVA OF *PENAEUS*

MYSIS LARVA OF *PENAEUS*

COMMENTS:

1. In **Penaeus**, the zoea larva, instead of converting into the megalopa stage, moults into the post larval mysis larva.
2. It is elongated, laterally compressed and transparent.
3. The body is divisible into **cephalothorax** and **abdomen**.
4. The cephalothorax is covered by a **carapace**.
5. The appendages are well developed. They include **5 pairs of cephalic appendages**,
8 pairs of thoracic appendages and **6 pairs of abdominal appendages**.
6. The thoracic appendages are used for swimming.
7. The eyes are **stalked** and **compound**.
8. The abdomen develops similar to that of the adult form, with 5 pairs of **biramous pleopods** and a pair of **uropods** and a **telson**.
9. The mysis larva metamorphosis in to the adult prawn by the loss of the exopodites on the thoracic legs.
10. The larval mysis differs from the adult mysis in the absence of a **brood pouch** beneath the thorax and a pair of statocysts.

:

AMPHIOXUS

(*Branchiostoma lanceolatum*)

SYSTEMATIC POSITION:

Phylum : Chordata
Subphylum : Cephalochordata
Class : **Leptocardii**
Order : **Amphioxiformes**
Genus : *Branchiostoma*
Species : *lanceolatum*

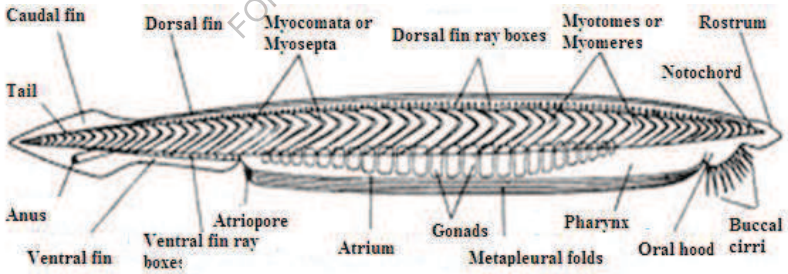


Fig.22. Amphioxus-*Branchiostoma lanceolatum*

AMPHIOXUS

(*Branchiostoma lanceolatum*)

COMMENTS:

1. **Amphioxus** is found in shallow subtidal waters (marine) with sandy bottoms. They reside in subtropical and temperate regions. Lancelets burrow their posterior end in the sand, with just their head exposed.
2. Branchiostoma is about 3.5 to 6.0 cm in length. The body is slender, somewhat translucent, superficially fish-like, laterally compressed and pointed at both ends, hence, the name lancelet.
3. The anterior two-thirds of the body is roughly triangular in transverse section. The posterior third is nearly oval in section. The pointed anterior end of trunk projects in front as a snout or rostrum. Below the rostrum is an oral hood formed by dorsal and lateral projections of the body. It bears twenty or more stiff oral cirri or tentacles.
4. The body is distinguished in two regions- the trunk and tail. The true head is lacking.
5. Branchiostoma has three unpaired fins- dorsal, caudal and ventral. Paired fins are absent. Along the mid-dorsal side of the body is a dorsal fin running along the whole length. It is joined to a somewhat broader caudal fin around the tail. Mid-ventrally is a ventral fin running from the caudal fin to an atriopore.
6. The dorsal fin is supported by a series of fin rays, and nerves the ventral fin by a series of paired fin rays. The fin rays are made of a gelatinous substance and connective tissue. Fin rays are lacking in caudal fin.
7. The oral hood is continuous posteriorly with two lateral membranous hollow metapleural folds running ventrally up to the atriopore.
8. The body wall shows metameric segmentation, being made of <-shaped blocks of muscles called myotomes or myomeres. Beneath the myotomes on both sides are present a series of gonads between mouth and atriopore.
9. The mouth surrounded by a frill-like membrane, the oral hood. Its edge is beset with numerous tentacles or cirri, immediately in front of the anterior end of the ventral fin, and partly enclosed by the metapleural folds, is a rounded aperture, the atriopore.
10. A short distance from the posterior extremity of the body (trunk) is the anus, placed asymmetrically on the left side at the base of the caudal fin. The post-anal portion of the body is called the tail.

SHARK

(*Scoliodon sorrakowah*)

SYSTEMATIC POSITION:

Phylum	: Chordata
Class	: Chondrichthyes
Subclass	: Elasmobranchii
Order	: Carcharhiniformes
Family	: Carcharhinidae
Genus	: <i>Scoliodon</i>
Species	: <i>sorrakowah</i>

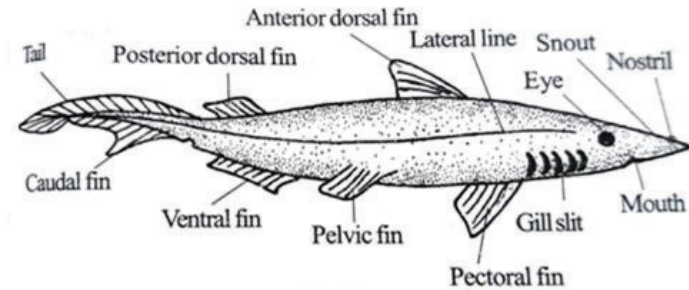


Fig 23. Shark (*Scoliodon sarrokowah*)

SHARK

(*Scoliodon sorrakowah*)

COMMENTS:

1. Shark is a marine, cartilaginous fish. The body boat shaped and divisible into three region namely head, trunk and tail.
2. Sharks do not have gas-filled swim bladders for buoyancy.
3. The head has a snout, a mouth a pair of nostrils a pair of eyes and five pair of gills slits.
4. The trunk has a pair of pelvic fins a dorsal fins and a cloacae. The tail has a dorsal fin a ventral fin and caudal fin the caudal fin is heterocercal.
5. The Lateral line sense organs two pairs are present it is a **rheoreceptors** to sense the salt level in sea water.
6. The shark skin is rough and the roughness of the skin is due to the presence of innumerable spine like structure called placoid scales.
7. The head and snout on the dorsal side contain numerous groups of pores called Ampullary pores. They are the external opening of **Ampulla of lorenzini**.
8. The respiratory system is formed of five pairs of gills pouches. They are located the external wall of pharynx.
9. The circulatory system consist of heart , blood, arteries and veins. The heart is located beneath the pharynx and covered by a sac called **pericardium**. Between the two layers of pericardial cavity filled with pericardial fluid.
10. The shark the sexes are separates. It exhibits sexual dimorphism.
11. The shark fertilization is Internal. The fertilized egg develops inside the uterus of the mother and the mother gives birth to young one. The embryo is nourished by the yolk stored in the egg and the mother gives mainly protection. This development is called **ovoviviparous**.

COBRA (*Naja naja*)

SYSTEMATIC POSITION:

Phylum	: Chordata
Class	: Reptilia
Order	: Squamata
Suborder	: Serpentes
Family	: Elapidae
Genus	: <i>Naja</i>
Species	: <i>N. naja</i>

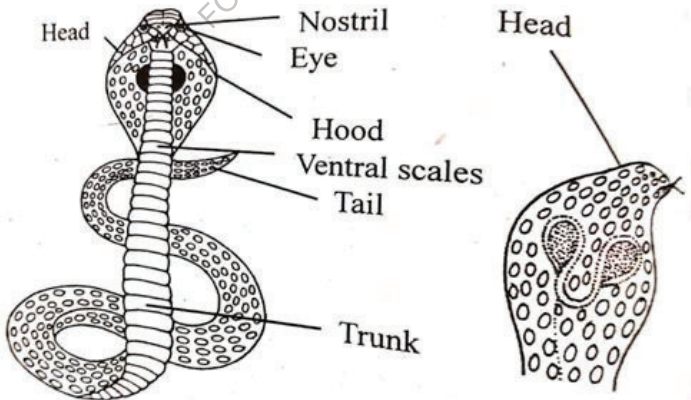


Fig 24. Cobra-*Naja naja*

COBRA

(*Naja naja*)

COMMENTS:

1. Cobra in burrows, deserted hills of termites, heaps of stones and stack of woods.
2. It is brown or black in colour and grows to a length of 6 feet.
3. It feeds on frogs, lizards, rats and small birds.
4. When distributed, it raises its head and spreads its neck as a hood. The hood is supported by cervical ribs. It always its head and hood to the right and left, backwards and forwards, makes a hissing sound in anger and gets ready to strike or bite.
5. The hood has a characteristic spectacle mark on the dorsal side and a pair of black patches on the ventral side.
6. The head is not distinct from the body.
7. The head is covered by shields. The third supralabial shield touches the eye and nostril.
8. It is highly poisonous. The poison is a **neurotoxin**. The poison of a single cobra can kill 15 persons at a time.
9. It has two poisonous gland and two fangs.
10. It cannot hear air borne sounds but can hear earth borne sounds.
11. The cobras are oviparous. The female lays 12-13 soft shelled eggs and the female incubates. A baby cobra, like all snakes, is called a **hatchling**. It is able to take care of itself from the start and can spread its hood and strike on the same day it hatches. A large yolk sac remains in the hatchling's stomach to give it nourishment for up to two weeks before it needs to find food on its own.

PIGEON

(*Columba livia*)

SYSTEMATIC POSITION:

Phylum : Chordata
Class : Aves
Order : Columbiformes
Family : Columboidae
Genus : *Columba*
Species : *livia*

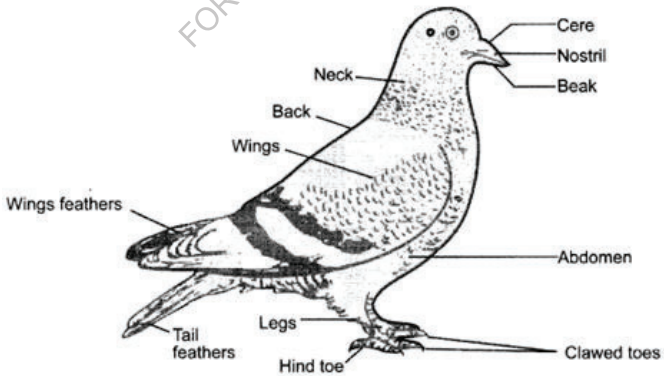


Fig. 24. Pigeon - External features

PIGEON

(*Columba livia*)

1. It is a slaty grey bird with glistening metallic green on the neck and the upper breast. It is found throughout India, Bangladesh, Pakistan, Srilanka and Burma.
2. They live in flocks and semi domesticated conditions and grain warehouse, railway stations and old buildings.
3. It feeds on cereals, pulses, groundnut etc.,
4. The body is spindle shaped and divisible into four regions namely **head, neck, trunk and tail**.
5. The body is covered by feathers. The feathers are four types namely **quill feathers, filoplumes, down feathers and rictal bristle**.
6. The digestive system consists of alimentary canal and digestive glands. It consists of mouth, buccal cavity, pharynx, oesophagus, stomach, intestine, rectum and cloaca.
7. The oesophagus leads to storage bag called **crop**. It consists of crop glands and secretes **pigeon's milk**. It is secreted by both the sexes. It has a more nutritive value than mammalian milk and nestlings are feed with by the parents.
8. The circulatory system of pigeon consists of blood, four chambered heart and covered with double layered protective membrane called **pericardium**. The layer is filled with the fluid is called pericardial fluid. The RBC is oval in shape and nucleated.
9. The pigeon exhibits **pulmonary** respiration. The respiratory system pigeon consists of lungs, air sacs air capillaries, bronchi, the air sac connected with the pneumatic activities of the bones.
10. In the tracheal region the sound is produced by **syrinx**. It is the sound box of birds.
11. Pigeon has well adaptation to move both in the air and on land.
12. The excretory system consists of a pair of kidney. The kidney are **metanephric** type.
13. The pigeon fertilization is internal, pigeon lays two eggs. The eggs are **cleidoic** and **telolecithal**.

BLASTULA OF FROG

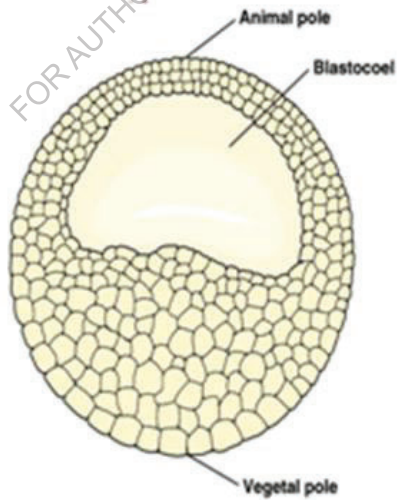
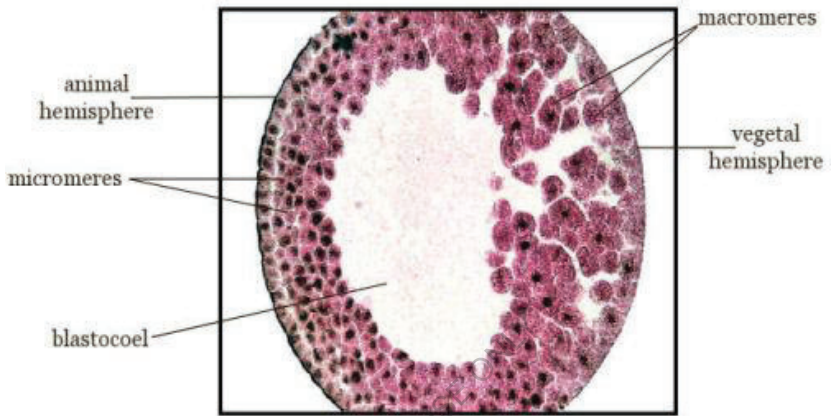


Fig. 25. Blastula of frog

BLASTULA OF FROG

Blastula is a hollow sphere of cells, referred to as blastomeres, surrounding an inner fluid-filled cavity called the blastocoele formed during an early stage of embryonic development in animals.

Embryo development begins with a sperm fertilizing an egg to become a zygote which undergoes many cleavages to develop into a ball of cells called a **morula**. Only when the blastocoele is formed does the early embryo become a blastula. The blastula precedes the formation of the gastrula in which the germ layers of the embryo form.

COMMENTS:

1. The first indication of gastrulation is the appearance of a small groove on the surface of the blastula just ventral to the grey crescent.
2. This slit is the beginning of the blastopore and represents the area where the cells forming the surface of the blastula move toward the inside to form the primary germ layers.
4. The section passes through the blastopore, which is visible as an indentation near the vegetal pole.
5. The dorsal lip of the blastopore can be seen as a finger like projection pointing toward the vegetal pole. Initial cell migration occurs at this site.
6. Slightly later in gastrulation, cells will begin to move inward over the ventral lip, and the blastopore will become a round opening. Movement of cells inward will result in the formation of the gastrocoele (or archenteron).
7. The archenteron will expand dorsally toward the animal pole obliterating blastocoel in the process.

24 HOURS CHICK EMBRYO

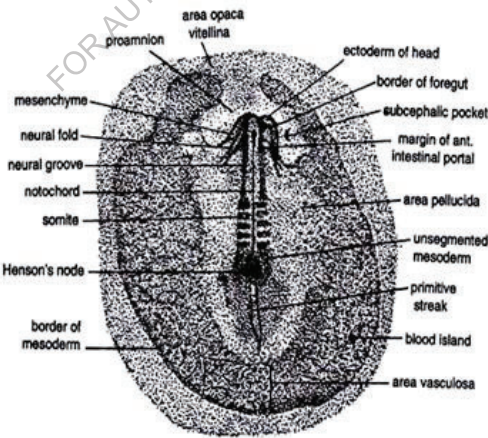
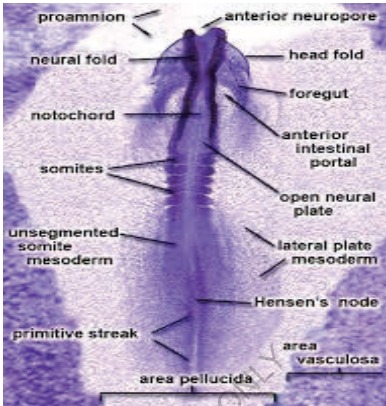


Fig. 26. Chick embryo-24 hrs incubation

24 HOURS CHICK EMBRYO

Note that in the 24 hour chick, Hansen's node is located further caudally and the primitive streak is present only at the posterior end of the embryo. As Hansen's node retreats posteriorly, the organs which define the embryonic axis are formed anterior to it. As the node retreats, the neural plate, underlain by the notochord, develops in front of it. Thus, neurulation and organ formation take place in a craniocaudal sequence.

1. The 24 hour chick embryo is a neural fold, which is ectoderm thickenings derived from the flattened neural plate.
2. The medial depression bordered by the neural folds is the **neural groove**. When the neural folds begin to fuse dorsally, they form the neural tube.
3. The neural tube forms the brain cranially and the dorsal hollow nerve tube caudally. The notochord is visible cranial to Hensen's node and between the neural folds. In the embryo, the notochord serves as a stiff, rod like supporting structure.
4. The first stage you have examined in which somites are present. Somites are blocks of mesoderm which appear on either side of the notochord.
5. The somites will give rise to a number of structures, including skeletal muscle, bone and the dermis of the skin. They first appear toward the anterior end of the embryo, and then progressively further posterior.
6. The first somite appears at about 21 hours and additional somites appear at the rate of about one per hour. Embryos are sometimes described by the number of somites, rather than the number of hours of incubation.
7. Note that blood islands are formed throughout the area opaque. Blood islands consist of irregularly shaped accumulations of mesodermal cells. The blood islands will form blood vessels and primitive blood cells.
8. At later stages they can be seen fusing together to form blood vessels.

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